(notes) 3.2 Domain and Range

Quick notes for Interval Notation

Using Words	Inequality	Interval Notation
All values of the variable z less than -3	$z > -3$	$(-3,\text{oo})$
All values of the variable z greater than or equal to -3	$z \geq -3$	$[-3,\text{oo})$
All values of the variable z between -3 and -2	$-3 < z < -2$	$(-3, -2)$
All values between -3 and -2, including -3	$-3 \leq z < -2$	$[-3, -2)$
All values less than -3	$z < -3$	$(\text{-oo}, -3)$

Included = solid dot = square bracket

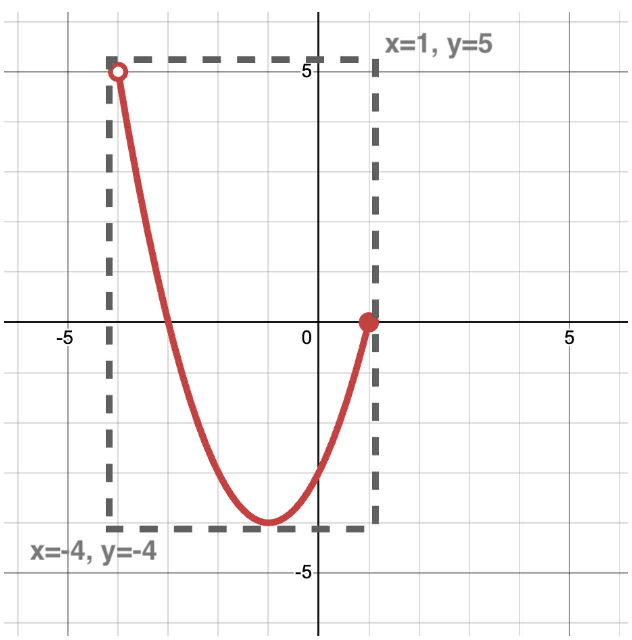
Excluded = open circle = parenthesis

Infinity

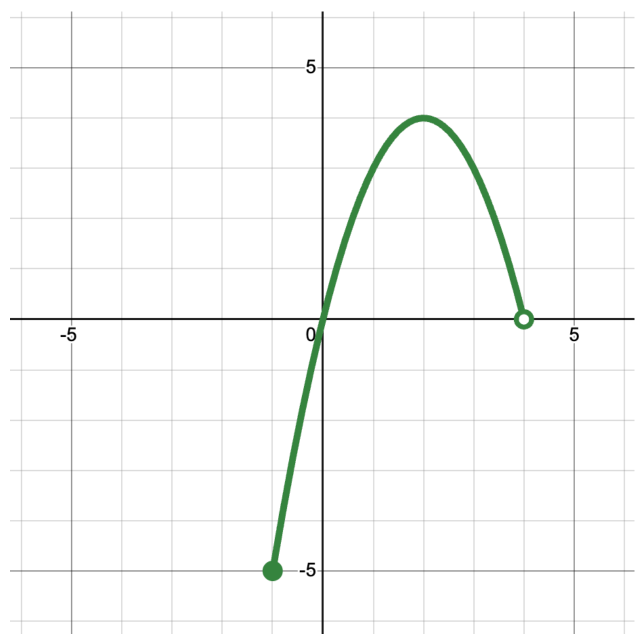
Negative infinity is always on the left side: $(-\infty, \ldots$

Positive infinity is always on the right side: $\ldots, \infty)$

Write the domain and range

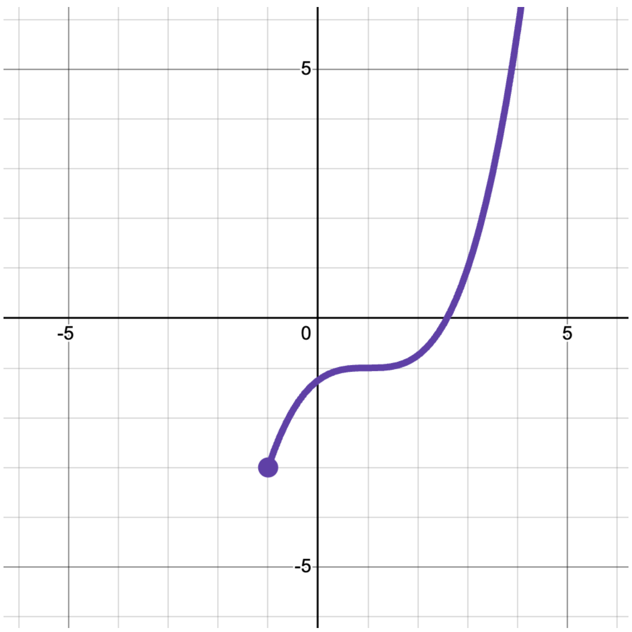


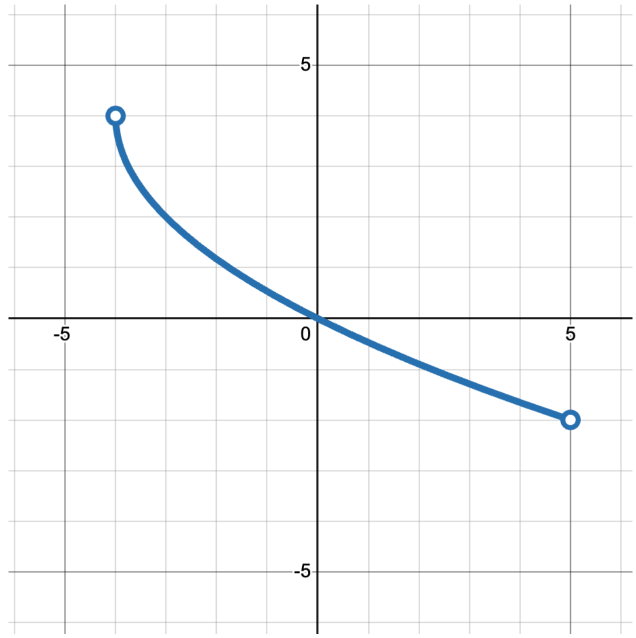
Make a rectangle around the graph and label the corners. Now we know the left most x value is x = -4 and the right most x value is x = 1. The domain is (-4, 1)
From the rectangle, we also see the bottom most part of the graph is y = -4 and the highest it goes is y = 5. The range is [-4, 5)



Domain:

Range:





Note that this graph continues off the screenshot to infinity!!

Domain:

Range:

Domain:

Range:

Domain Restrictions

Try it first without a calculator and then do it with a calculator:

$\frac{7}{0} =$ $\frac{0}{0} =$ $\frac{0}{7} =$

$\sqrt{4} =$ $\sqrt{-4} =$ $\frac{1}{\sqrt{4}} =$

Question: Did all your calculations work?? And if not, what kind of error did you get?

When to beware: fractions and square roots!

When x is in the denominator, we must restrict the domain to exclude the x's that will cause the denominator to become zero. When we divide by zero, the value is __________

ex1. $f(x) = \frac{1}{x-2}$

To find the restriction, set the denominator equal to zero and solve for x

$$x - 2 \neq 0$$
$$x - 2 + 2 \neq 2$$
$$x \neq 2$$

Domain: $\{x | x \neq 2\}$

ex2. $f(x) = \frac{x+1}{x+6}$

There are x in the numerator and denominator, just focus on the denominator

$$x + 6 \neq 0$$
$$x + 6 - 6 \neq -6$$
$$x \neq -6$$

Domain: $\{x | x \neq -6\}$

When x is under an even radical (ex. $f(x) = \sqrt{x},\ \text{or}\ f(x) = 2\sqrt{x+7}$), we must restrict the domain to ensure the value under the radical is greater than or equal to zero. If there is a negative number under the radical, the value is ______________.

ex3. $f(x) = \sqrt{x-8}$

To find the domain, set under the radical to greater than or equal to x

$$x - 8 \geq 0$$
$$x - 8 + 8 \geq 8$$
$$x \geq 8$$

Domain: $\{x|x \geq 8\}$

ex4. $f(x) = -5\sqrt{-x-3}$

Set the expression under the radical to greater than or equal to x. Don't worry about outside the radical

$$-x - 3 \geq 0$$
$$-x - 3 + 3 \geq 3$$
$$-x \geq 3$$
$$x \leq -3$$

Domain: $\{x|x \leq -3\}$

Find the domain restrictions for the following functions:

$f(x) = \frac{x-5}{x+3}$

Domain restriction:

$f(x) = -3\sqrt{5x-17}$

Domain restriction:

$f(x) = 5\sqrt{x+7}$

Domain restriction:

$f(x) = \frac{x^2-9}{x^2+7x+10}$

Hint: look at the graph for this one!

Domain restriction:

^view the graphs on desmos